VRAJ PATEL

AI / Applied AI Engineer

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PROFESSIONAL SUMMARY

AI / Applied AI Engineer with 2+ years of experience building production-oriented AI systems across Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), agentic AI, computer vision, OCR, and workflow automation. Hands-on with Python, FastAPI, Flask, LangChain, LangGraph, Pinecone, Qwen fine-tuning, LLM evaluation, guardrails, Docker, and Google Cloud Run. Delivered measurable outcomes including 95% OCR extraction accuracy, 15% detection precision improvement, 40% faster knowledge retrieval, 50%+ hiring-time reduction, and AI backend workflows handling 200k-250k requests/sec.

TECHNICAL SKILLS

LLM / GenAI: Large Language Models (LLMs), Generative AI, Prompt Engineering, Function Calling, Tool Calling, Structured Outputs, Context Engineering, Context Assembly, LLM Inference, LLM Serving, Model Context Protocol (MCP), Guardrails
Agentic AI: Agentic AI, Multi-Agent Systems, AI Agents, Agent Orchestration, Agent Memory, Stateful Agents, Human-in-the-Loop (HITL), Tool Connectors, Secure Action Execution, LangGraph, LangChain
RAG / Retrieval: Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, Semantic Search, Hybrid Search, Reranking, Chunking Strategies, Retrieval Evaluation, Citation-Aware Retrieval, Knowledge Retrieval, Pinecone, FAISS
AI Evaluation / LLMOps: LLM Evaluation, Agent Evaluation, Model Evaluation, Evaluation Harnesses, LLM-as-Judge, Golden Datasets, Regression Test Suites, Observability, Tracing, MLflow
Machine Learning / Data Science: Machine Learning, Deep Learning, NLP, Computer Vision, OCR, YOLOv8, Face Recognition, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Feature Engineering, Exploratory Data Analysis (EDA), Classification, Regression, Statistical Modeling, Time-Series Forecasting
Programming / Data: Python, Pandas, NumPy, SciPy, SQL, MySQL, SQLite, MongoDB, Data Cleaning, Data Processing, Statistical Analysis, Data Visualization
Backend / APIs: FastAPI, Flask, REST APIs, API Development, Microservices, API Security, Async Processing, Async Job Queues, Concurrent Workflows, WebSockets, Streaming APIs, Video Streaming, Streamlit
MLOps / Cloud / DevOps: Docker, CI/CD, Google Cloud Run, GCP, AWS, Model Deployment, Model Serving, Autoscaling, Experiment Tracking, Monitoring, Scalability, Latency Optimization, Production Reliability, Git
System Design / Production AI: System Design, Distributed Systems, Latency/Throughput Trade-Offs, Caching, Rate Limiting, Fault Tolerance, Production AI Workflows
ERP: Microsoft Dynamics 365 Business Central, AL, RDLC Reports, ERP Customization

PROFESSIONAL EXPERIENCE

AI Engineer | Data Scientist

Hubblehox Technologies | Ampersand Group
July 2024 - Present

- Built OCR and computer-vision pipelines using YOLOv8 region detection, preprocessing, feature engineering, hyperparameter tuning, and Google OCR, achieving 95% data-extraction accuracy and improving detection precision by 15%.
- Developed a Flask-based AI interview automation platform with resume/JD parsing, LLM-powered question generation, structured outputs, candidate evaluation, async job handling, video/audio workflows, and TTS/Whisper automation, reducing teacher hiring time by 50%+.
- Built a RAG-based internal AI assistant using embeddings, chunking strategies, Pinecone vector search, semantic search, retrieval evaluation, and context assembly for grounded Q&A, improving internal query response performance by 40%.
- Implemented AI proctoring workflows using head-pose detection, eye tracking, audio/noise analysis, video-streaming inputs, classification-style signal checks, and optimized open-source computer-vision models for Maharashtra government examinations.
- Created domain-specific datasets and fine-tuned Qwen 1.7B for PCM/PCB question generation using dataset engineering, SFT, prompt/response curation, evaluation harnesses, regression-style quality checks, and post-training workflows.
- Engineered production-oriented agentic AI workflows with agent orchestration, tool/function calling, retries, guardrails, stateful workflow handling, structured response validation, observability/tracing, distributed-system design, rate limiting, caching, latency/throughput optimization, fault-tolerant execution, Cloud Run model serving, autoscaling, and 200k-250k requests/sec throughput.

AI Data Trainer / LLM Evaluator

Freelancing | Remote

- Evaluated Claude and Gemini outputs across reasoning, coding, debugging, code review, factuality, safety, completeness, and instruction following.
- Reviewed coding-agent workflows for repo understanding, tool use, code edits, tests, bug fixing, error recovery, task completion, maintainability, and security risks.

Microsoft Business Central Technical Consultant

QuantumQuad Solutions
January 2024 - June 2024

- Customized Microsoft Dynamics 365 Business Central modules using AL extensions, page/table objects, validation rules, approval flows, and RDLC/Word reports.
- Supported ERP implementation, APIs, import/export routines, configuration, troubleshooting, UAT, production issue resolution, and client workflow mapping.

Machine Learning Intern

Languify | Live Face Recognition Project
August 2022 - September 2022

- Supported a live face-recognition ML project involving data cleaning, preprocessing, integration, statistical analysis, visualization, and model-development workflows.

SELECTED AI/ML PROJECTS

LawyerAI - Multi-Agent Legal Research and Case Advisory Platform

Python | FastAPI | Angular | MongoDB | LangGraph | RAG | Multi-Agent AI | LegalTech
github.com/iamvrajpatel/lawyer-ai

- **Used Skill:** FastAPI, Angular, MongoDB, LangGraph, RAG, multi-agent systems, legal agents, knowledgebases, evidence files, audit logs.
- Built a case-centric legal AI workspace with editable cases, multi-session chat, evidence management, scoped retrieval, schema validation, legal disclaimers, and configurable agent flows for intake, jurisdiction, evidence review, risk/compliance, negotiation, loophole analysis, and response synthesis.

Qwen Fine-Tuning for PCM Entrance Exam Question Generation

Python | Qwen3-1.7B | QLoRA | LoRA | TRL SFTTrainer | FastAPI | SQLite | Dataset Engineering
github.com/iamvrajpatel/qwen-finetune-q-gen

- **Used Skill:** Qwen3-1.7B, QLoRA, LoRA, TRL SFTTrainer, 4-bit NF4, FastAPI, SQLite, JSONL, NCERT, Bloom taxonomy.
- Built a source-grounded dataset-prep and fine-tuning pipeline with NCERT retrieval, distribution scheduling, semantic validation, duplicate rejection, JSONL/CSV export, train/eval splits, LoRA rank 8, gradient accumulation, and structured JSON question outputs.

Lip-Sync and Voice-Cloning Deepfake Video Generation API

Python | FastAPI | Gradio | PyTorch | Diffusers | LatentSync | LivePortrait | Voice/Audio Processing | OpenCV | FFmpeg
github.com/iamvrajpatel/Lip-Sync

- **Used Skill:** LatentSync, LivePortrait, PyTorch, Diffusers, FastAPI, Gradio, FFmpeg, OpenCV, torchaudio, SyncNet, FVD, CUDA.
- Built a lip-sync/deepfake API for image/video plus target voice audio with async jobs, uploads, polling, downloads, CORS, cleanup, LivePortrait reference-video generation, FFmpeg preprocessing, FPS conversion, audio padding, inference controls, and CUDA cleanup.

AI Interview Automation Platform

Python | Flask | OpenAI | Whisper | FaceNet | MTCNN | TTS | NLP | Audio/Video Processing
github.com/iamvrajpatel/Interview-Automation

- **Used Skill:** Flask, OpenAI, Whisper, FaceNet, MTCNN, TTS, resume parsing, JD parsing, scoring, recommendation engine.
- Built an end-to-end interview workflow with face verification, resume/JD upload, adaptive question generation, webcam/audio recording, Whisper transcription, answer scoring, topic extraction, results dashboard, and recommendation generation.

Face Verification API Using Open-Source Models

Python | FastAPI | DeepFace | FaceNet | MTCNN | RetinaFace | TensorFlow | OpenCV | REST API
github.com/iamvrajpatel/Face-Authentication-API

- **Used Skill:** FastAPI, DeepFace, FaceNet, MTCNN, RetinaFace, TensorFlow, Keras, OpenCV, REST API, similarity scoring.
- Built a face-verification endpoint comparing registered/test image URLs with match status, confidence percentage, configurable threshold behavior, JSON contracts, validation, HTTP errors, and reusable identity verification logic.

EDUCATION

WorldQuant University: M.S. Financial Engineering, June 2025 - Expected July 2027
G H Patel College of Engineering and Technology, Anand: B.E. Computer Engineering, September 2020 - May 2024 | CPI: 8.2 / 10.0

CERTIFICATIONS

- IBM Data Science Specialization
- Machine Learning for Trading Specialization
- Vector Databases Professional Certificate - Weaviate - Deeplearning.AI
- Automated Testing for LLMOps - Deeplearning.AI
- Evaluating AI Agents - Deeplearning.AI
- Fine-Tuning & RL for LLMs : Intro to Post Training - Deeplearning.AI
- Agentic AI - Deeplearning.AI